Before we formally introduce this, we require the notion of computation tree and introduce some more notations.

Definition 1: [16] A *computation tree* corresponding to a message-passing decoder of the Tanner graph G is a tree that is constructed by choosing an arbitrary variable node in G as its root and then recursively adding edges and leaf nodes to the tree that correspond to the messages passed in the decoder up to a certain number of iterations. For each vertex that is added to the tree, the corresponding node update function in G is also copied.

Let G be the Tanner graph of a 3-left-regular code. Let H be the induced subgraph of a trapping set (a, b) contained in G with variable node set $P \subseteq V$ and check node set $W \subseteq C$. Let $\mathcal{N}(u)$ denote the set of neighbors of a node u. Let $\mathcal{T}_i^k(G)$ be the computation tree of graph G corresponding to a decoder $\mathscr{F}$ enumerated for k iterations with variable node $v_i \in V$ as its root. Let $W' \subseteq W$ denote the set of degree-one check nodes in the subgraph H. Let $P' \subseteq P$ denote the set of variable nodes in H where each variable node has at least one neighbor in W'. During decoding on G, for a node $v_i \in P'$, let μ_l denote the message that v_i receives from its neighboring degree-one check node in H in the l^{th} iteration.

Definition 2: A vertex $w \in \mathcal{T}_i^k(G)$ is said to be a *descendant* of a vertex $u \in \mathcal{T}_i^k(G)$ if there exists a path starting from vertex w to the root v_i that traverses through vertex u. The set of all descendants of the vertex u in $\mathcal{T}_i^k(G)$ is denoted as $\mathcal{D}(u)$. For a given vertex set U, $\mathcal{D}(U)$ (with some abuse of notation) denotes the set of descendants of all $u \in U$.

Definition 3: $\mathcal{T}_i^k(H)$ is called the computation tree of the subgraph H enumerated for k iterations for the decoder $\mathscr{F}$, if $\forall\ c_j \in W'$, μ_l is given for all $l \leq k$, and if the root node $v_i \in P$ requires only the messages computed by the nodes in H and μ_l to compute its binary hard-decision value.

Definition 4 (Isolation assumption): The computation tree $\mathcal{T}_i^k(G)$ with the root $v_i \in P$ is said to be isolated if: (i) for any check node $c_j \in W'$ that is in $\mathcal{T}_i^k(G)$ with $v_t \in P'$ as its parent, $\mathcal{D}(c_j) \cap \mathcal{D}(\mathcal{N}(v_t) \setminus c_j) = \emptyset$, and (ii) for any two check nodes $c_r, c_s \in W \setminus W'$ that are also in $\mathcal{T}_i^k(G)$, $\mathcal{D}(c_r) \cap \mathcal{D}(c_s) \subseteq (P \cup W)$. If $\mathcal{T}_i^k(G)$ is isolated $\forall v_i \in P$, then the subgraph H is said to satisfy the isolation assumption in G for k iterations.

Remark: The isolation assumption can still be satisfied even when there are nodes in H that appear multiple times in $\mathcal{T}_i^k(G)$ as long as these nodes are not descendants of the degree-one check nodes. Whereas Gallager's independence assumption will be violated if any node in H is repeated in $\mathcal{T}_i^k(G)$. Hence, isolation assumption is a weaker condition than independence. For clarity, we illustrate with an example shown in Fig. 1.

Example 1: Let us assume that the graph G of code $\mathcal{C}$ contains a subgraph H induced by a $(6, 2)$ trapping set. Fig. 1 shows the subgraph H, and the computation tree $\mathcal{T}_3^2(G)$ of graph G with v_3 as its root enumerated for two iterations. The ■ denotes a odd-degree check node. The solid lines represent connections within subgraph H and the dotted lines represent connections from the rest of the graph G outside the subgraph

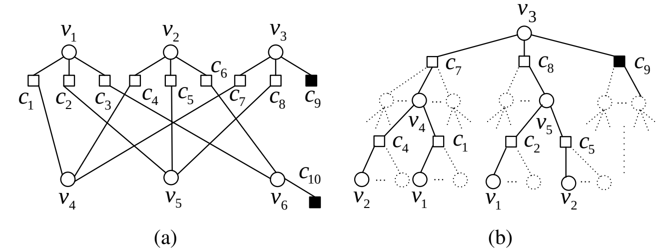

Fig. 1. Subgraph H induced by (6,2) trapping set contained in G: (a) Tanner graph of H; (b) computational tree $\mathcal{T}_3^2(G)$

H. The isolation assumption is satisfied for two iterations if none of the descendants of the check nodes c_7 and c_8 appear as a descendant of check node c_9 (similar condition has to hold for c_{10}), and if the only common descendants of the degree-2 check nodes are nodes in H. But the independence assumption does not hold for two iterations.

Theorem 1 (Isolation theorem): Let $G = \{V \cup C, E\}$ of a 3-left-regular code which contains a subgraph $H = \{P \cup W, E'\}$ that is induced by a trapping set (a, b). Let $W' \subseteq W$ denote the set of degree-one check nodes in H and let $P' \subseteq P$ denote the set of variable nodes in H where each has at least one neighbor in W'. If $\mathbf{r}$ is input to decoder $\mathscr{F}$ from the BSC such that $\text{supp}(\mathbf{r}) \in P$, and if H satisfies the isolation assumption in G for k iterations, then for each $c_j \in W'$, the message from c_j to its neighbor in H in the l^{th} iteration denoted by μ_l, is determined as the output of $\Phi_v(\mu_{l-1}, \mu_{l-1}, \text{C})\ \forall l \leq k$.

Proof: This follows by looking at the computation tree $\mathcal{T}_i^l(G)$ where $l \leq k$ with any $v_i \in P'$ as its root. Let the initial messages passed from a variable node be $\pm\mu_0 \in \mathcal{M}$. Since $\text{supp}(\mathbf{r}) \in P$, due to the isolation assumption, this means that all the variable nodes that are descendants to any $c_j \in W'$ are initially correct. In the initial iteration, from the definition of Φ_c, the outgoing message of check nodes that are descendants to c_j is μ_0. In the next iteration, the variable nodes connected to these check nodes receive μ_0 on all their edges due to the isolation assumption and send $\Phi_v(\mu_0, \mu_0, \text{C})$ as their outgoing messages. Due to the definition of Φ_c, the check nodes connected to these nodes in $\mathcal{T}_i^l(G)$ send μ_1 which is simply $\Phi_v(\mu_0, \mu_0, \text{C})$. This process inductively follows while traversing up the tree for l iterations. Moreover, computation of the hard-decision value for any node v_i in H requires only messages from nodes in H in addition to μ_l. ■

Remark: Note that the isolation assumption and theorem can be restated for the min-sum decoder.

Corollary 1: Consider the min-sum decoder for 3-left-regular LDPC codes with $\mathcal{Y} = \{\pm 1\}$. If subgraph H contained in G satisfies the isolation assumption for k iterations, and if all variable nodes outside H are initially correct, then μ_l of the degree-one check node for the min-sum decoder is $2\mu_{l-1} + 1$.

Corollary 2: If H is a subgraph contained in G such that it satisfies the isolation assumption for k iterations, and if all variable nodes outside H are initially correct, then the computation tree $\mathcal{T}_i^k(G)$ with $v_i \in P$ is equivalent to $\mathcal{T}_i^k(H)$, provided μ_l for each degree-one check node in H is computed